Reg. No. | | | | | | | | | | [18MTEST231]

SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

REAL TIME EMBEDDED SYSTEMS

Time: 03 Hours **Maximum Marks:100**

Instructions:

1. Answer **FIVE** full Questions. Question1 and Question2 are compulsory.
2. Missing data may be suitably assumed.

1.	**a.**	What are real time systems? What are their characteristics? Explain different types of real time systems with an example.	**10**
	b.	Define hard real-time services and soft real-time services. Describe why and how they are different.	**10**
2.	**a.**	Describe why a multilevel cache architecture might not be ideal for hard real-time embedded systems.	**10**
	b.	Write short note on: **i.** Shared memory **ii.** Flash file systems.	**10**
3.	**a.**	Explain the way of handling missed deadlines and Quality of Service (QoS) for a real time system.	**10**
	b.	Explain with example various hardware and firmware components.	**10**
		OR	
4	**a.**	Discuss two algorithms for the determination of necessary and sufficient feasibility testing with rate monotonic policy.	**10**
	b.	Write short notes on the various communicating and synchronized applications.	**10**
5.	**a.**	Explain the example stereo vision UML Visio template design provided on CD-ROM.	**10**
	b.	Describe whether CVS is better than subversion or vice versa and why.	**10**
		OR	
6.	**a.**	Explain spiral process of systems engineering with emphasis on high level design.	**10**
	b.	Explain the full-duplex VoIP data flow with a neat block diagram.	**10**
7.	**a.**	Build a simple single change relay or MOSFET H-bridge reversible motor controller and design a serial or parallel port command interface for it.	**10**
	b.	Implement a stereo imaging system that uses a laser pointer to paint a three-dimensional scene and create a three-dimensional map of the area.	**10**
		OR	
8.	**a.**	Using a C-program or MATLAB model, develop a kinematics model for the reach capability of the OWI five-degree-of-freedom robotic arm.	**10**
	b.	Implement a stereo range finder that can also tilt/pan track a moving object.	**10**

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